INCH-POUND

MIL-S-44041C(GL)
9 October 1990
SUPERSEDING
MIL-S-44041B(GL)
30 June 1988

MILITARY SPECIFICATION

SHIRT, MAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a polyester/cotton broadcloth durable press shirt.

1.2 Classification. The shirt shall be of the following sizes as specified (see 6.2).

Schedule of sizes - inches

13	16
13-1/2	16-1/2
14	17
14-1/2	17-1/2
15	18
15-1/2	

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8405

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documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester-Covered
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-44193	-	Thread, Polyester, Textured
MIL-T-43624	-	Thread, Polyester, Spun
MIL-C-43992	-	Cloth, Broadcloth, Polyester/Cotton, Durable Press
MIL-C-44296	-	Cloth, Fusibles
MIL-P-55010	-	Plastic Sheet, Polyethylene Terephthalate

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-1492	-	Provisions for Evaluating Quality of Men's Shirts

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Technical Manual of the American Association of Textile Chemists and Colorists

Method 88B (Two photographs) - Photographic Comparative Rating of Single and Double Needle Seams

Method 124-78 - Appearance of Durable Press Fabrics after Repeated Home Launderings

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709-2215.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type Number 304 - Stainless and Heat Resistant Steel
Type Number 316 - Stainless and Heat Resistant Steel

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 204 - Methods of Testing Sewing Threads
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

Department of Defense Standard Shades for Buttons

(Application for color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992. When cloth woven on shuttleless looms is used, the selvage for all types (i.e., leno, frayed, monofilament) shall not be used as part of the shirt and must be removed prior to shirt fabrication.

3.3.1.1 Button stays. The button stay material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992 or as an alternate, a commercial button stay material may be used.

3.3.2 Fusible interlining.

3.3.2.1 Shoulder loop, pocket flaps and front facing. The fusible interlining material for the shoulder loops, pocket flaps and front facing shall conform to type V, style B, class 2 of MIL-C-44296.

3.3.2.2 Collar and collarstand. The fusible interlining for the collar and collarstand shall conform to type I, style D, class 3 of MIL-C-44296. As an alternate, the fusible interlining as described in 3.3.2.1 may be used.

NOTE: Collar and collarstand fusible interlining shall always be of the same type.

3.3.2.2.1 Bonding strength. When tested as specified the fusibles in 3.3.2.1 and 3.3.2.2 shall meet the initial and laundered bonding strength requirements as specified in MIL-C-44296 when fused to the basic material.

3.3.3 Labels. Each shirt shall have an identification label, size label and instruction label conforming to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

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3.3.3.1 Identification label. The identification label shall conform to type VI, class 1.

3.3.3.2 Size label. The size label shall conform to type IV, class 2.

3.3.3.3 Instruction label. The instruction label shall conform to type VI, class 3, except that the label cloth shall be dyed Lemon Yellow, Cable No. 70205, in conformance with the Standard Color Card of America. The label shall contain the following information:

SHIRT, MAN'S, SHORT SLEEVE POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

Washing instructions

1. Wash with moderately hot water (115°F) using soap or mild detergent.

   a. For machine washing. Use wash and wear cycle.
   b. For hand washing. Do not wring or twist.

2. DO NOT STARCH, BLEACH OR USE FABRIC SOFTENER.

3. Dry in tumble dryer at warm setting. Remove immediately at end of drying. Shape and hang.

4. To drip dry, remove from water before last spin cycle. Shape and hang.

5. If touch up is required, iron at a moderate or synthetic material heat setting.

DO NOT REMOVE THIS LABEL

3.3.3.4 Combination identification-instruction label. The identification label and instruction label may be combined into a type VI, class 15 label, except the 1/2 inch space between the labels may be reduced to 1/4 inch. When combined, the color of the label shall conform to the requirements of the instruction label (see 3.3.3.3) and the item nomenclature shall not be repeated. The coating requirements of the type VI, class 3 label shall apply.

3.3.4 Thread. The thread for seaming, stitching, buttonholes and button sewing shall be cotton- or polyester-covered polyester core, ticket Nos. 70, 2 or 3 ply and 100, 2 or 3 ply conforming to A-A-50199. Spun polyester thread conforming to class 1 or 2 of MIL-T-43624 may be substituted for the cotton-covered polyester core thread as shown below. The thread shall be stabilized so as not to shrink in boiling water more than 1.5 percent when tested as specified in 4.4.1.1.

Spun polyester	Polyester-covered or cotton-covered polyester core
70, 2 or 3 ply	70, 2 or 3 ply
100, 2 ply	100, 2 or 3 ply

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As an alternate the looper thread for overedging the front facing fusible interlining to the front edge of the basic material may be polyester textured, Type I, class 2, size 18 Tex, conforming to MIL-T-44193.

3.3.4.1 Thread color and colorfastness. The thread shall be White AH, C.A. 66050. The dyed thread shall show fasteness to light, perspiration, and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light, perspiration, and laundering.

3.3.5 Buttons. The buttons shall conform to type II, class K, style 25, 19 line of V-B-871. The color of the buttons shall be shade White BA, C.A. 62031.

3.3.6 Collar stays. The collar stays shall be made from polyethylene terephthalate plastic sheet conforming to MIL-P-55010, except the thickness shall be 0.014 ± 0.001 inch, the color shall be natural (clear or opaque), and the transparency and optical properties shall not apply. Either both ends shall be rounded or pointed with the points rounded, or one end rounded and one end cut at any angle with the point rounded. Examination shall be as specified in 4.4.1.2.

3.3.6.1 Collar stay configurations.

3.3.6.1.1 Unslotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 2-3/4 to 2-7/8 inches long.

3.3.6.1.2 Slotted stays. The stays shall measure 1/2 ± 1/32 inch wide and 2-3/4 to 2-7/8 inches long. The stay shall have two slots 1-3/8 ± 1/32 inches long, centered along the length of the stay. Each slot shall be 1/32 ± 1/64 inch wide and spaced approximately an equal distance across the width of the stay.

3.4 Design. The shirt is a short sleeve dress style with a convertible collar. The shirt has a seven button front, two breast pockets, with button through flaps, pen pocket inside the left breast pocket and shoulder loops. The collar, collarstand, shoulder loops, pocket flaps and front facing have a fusible interlining; the collars have stays; and the shirt is of a safety-stitch construction with a yoke. The front facing is fused to the front shirt body (see figure 1).

3.5 Patterns. Standard patterns will be furnished by the Government and, unless otherwise indicated, provide an allowance of 1/4 inch for all seams except 3/8 inch for setting the pen and breast pockets. The standard patterns shall not be altered in any way (except when automated apparel equipment is used for either making or setting pen and breast pockets and/or flaps and shoulder loops, see 3.9) and are to be used only as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns.

3.5.1 List of pattern parts. The components of the shirt shall be cut from the material as specified below, in accordance with the pattern parts indicated.

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TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Basic material	Front	2
	Back	1
	Yoke	2
	Sleeve	2
	Top shoulder loop	2
	Under shoulder loop	2
	Top collar	1
	Under collar	1
	Pocket 2/	2
	Pocket flap 2/	4
	Pen pocket	1
	Stay holder 1/	2
Fusible interlining	Shoulder loop interlining	2
	Pocket flap interlining	2
	Top collarstand interlining	1
	Collar interlining	1
	Front facing interlining	2
Template	Pocket marker	

1/ If used.

2/ 1/8 inch may be added to three sides (see operation 1.e.).

3.6 End item seam appearance, bond strength of fusibles and adhesion of ultrasonically fused collar stays.

3.6.1 Seam appearance rating. After the finished shirt has been laundered five times (see 4.4.1.1). The appearance rating of the seams and stitching shall be as follows:

Parts	Minimum average of 3 observations
Fused collar	3.8
Shoulder loops	3.3
Pocket flaps	3.3
Pockets	2.8
Armholes	2.8
Sleeve hems	2.8
Side seams	2.8
Underarm seams	2.8
Yoke	2.8

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3.6.2 Bond strength of fusibles. After the shirt has been laundered 10 times, the shirt shall show no bubbling or delamination of the fused plies of material when tested as specified in 4.4.1.1. There shall be no showing of the "strike-through" of the adhesive after fusing.

3.6.3 Adhesion of ultrasonically fused collar stays. After the shirt has been laundered 10 times, the shirt shall show no delamination or curling of the ultrasonically fused collar stays when tested as specified in 4.4.1.1. There shall be no delamination of any fused points. (See 6.8 for recommended ultrasonic welding conditions).

3.7 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.7.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.7.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/4 to 1/2 inch except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in puckering of material sewn. The lock shall be embedded in the material sewn.

3.7.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch behind the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1 inch before the defective area and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.7.1.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of seams or stitching produced with a 401 stitch type, when not turned under a hem or held down by other stitching, shall

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have a 1/2 to 3/4 inch chain extending beyond each end. The looper (under thread) shall always be on the inside of the shirt. Repairs may be made by using a 301 stitch in accordance with 3.7.1.1.1a or 3.7.1.1.1b.

3.7.1.3 Type 502, 503, 515, 516 and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.7.1.1.1.a and b except substitute 3/4 inch for 1 inch wherever 1 inch appears. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn.

3.7.1.4 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches (minimum). Bartacks shall be free from thread breaks and loose stitching.

3.7.2 Buttonholes. The buttonholes shall be straight-cut, purl or whip type and the finished cut length shall be 1/2 to 9/16 inch long. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. When a purl type buttonhole is used, the purling shall be on the outside of the shirt, except for the neck closure buttonhole which shall have the purling on the inside of the shirt. All loose thread ends shall be trimmed or fused to within 1/8 inch of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric prior to stitching the buttonholes are prohibited.

3.7.3 Fusing. A single layer of fusible interlining shall be fused to a single layer of basic material, on an electric dry heated conveyor or flathead type fusing press which has the capability of controlling and retaining pressure, dwell time, and temperature within an 8-hour period. For optimum results with a particular fusible style and press used, setting should be obtained from the manufacturer of the fusible interlining.

3.7.4 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and at least three or more tying, overlapping or backstitches are used to secure the ends of stitching.

3.7.4.1 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

3.7.5 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirts.

3.8 Manufacturing operations requirements. The shirt shall be made in accordance with operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed unless otherwise

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specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread is removed or does not show on the finished shirt.

3.9 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table II, providing that the seam and stitch type are as specified and the finished component conforms to the required size, dimensions, and configurations. When a government finished shaper pattern is forwarded, caution shall be taken to assure the component conforms to that pattern. Patterns may be altered to accommodate the use of automated apparel equipment as specified in 3.5.

3.10 Figures. Figure 1 is furnished solely for guidance and information. Should variation from the written specification appear in figure 1, the written specification shall govern.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut shirts in strict accordance with patterns furnished which show size, directional lines, marks, and notches for proper assembly. Directional lines shall be placed in the warp direction. The fronts shall be cut on the selvage. When fronts of shirts are cut from the uneven edge of the lay, they shall be blocked out and relaid so that the width of the facings shall be exactly as the pattern. When selvageless basic cloth is utilized (i.e., cloth woven on shuttleless looms, see 3.3.1), the facings shall be overedge stitched as indicated in operation 9., and blocking will not be required. The use of drill holes are prohibited, except for the pencil pockets and sewn labels, providing the drill holes are covered by the outer fabrics or label respectively, otherwise exposed drill holes shall be considered to be holes in the fabric. b. Cut component parts from the same piece of the basic material except the undercollar, stay holders, button stay, pen pocket, the under parts of the pocket flaps, shoulder loops, and yoke which may be cut from ends. Parts cuts from ends shall approximate the shade of the other parts. c. Cut the fusible interlining in accordance with directional lines on patterns. d. Cut button stay from basic material 3/4 + 1/8 inch diameter or square.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) e. The pocket and flap may be cut 1/8 to 3/16 inch larger on three sides (excluding top of pocket and top of flap) however, the finished dimensions must be maintained.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1492 (see 4.4.3) shall be removed from production and replaced with non-defective and properly matched components.						
3.	Component marking. a. Mark, ticket all components of the shirt except those parts cut from ends to insure correct size, proper assembly and a uniform shade throughout the shirt. b. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the shirt and the numbers are covered by the seam allowance. c. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew on type tickets. 3. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labeling.						
	a. Center size marking on top collar stand (1/4 inch off center tolerance), so that on the finished shirt, the bottom of the marking will be not less than 1/4 inch or more than 1/2 inch from the seam joining the collar stand to the neck.						
	b. Position the yellow instruction label on the outside right front (as worn) so that on the finished shirt the bottom of the label shall be 1 to 2 inches from the hemmed bottom and the side of the label shall be 2-1/4 to 3 inches from the front folded edge and stitch on all four sides. The stitching shall not be through the printing.	301	LSbj-1	8-14	70-2	70 2	
	c. Position and stitch the identification label as in "b" above except it shall be placed on the left front.	301	LSbj-1	8-14	70-2	70-2	
	d. The yellow combination label, when applicable, shall be positioned and stitched as in "b" above.	301	LSbj-1	8-14	70-2	70-2	
5.	Make collar.						
	a. Unslotted collar stay holder construction - (when used).						
	(1) Position fusible interlining to top collar 1/16 inch from top and side edges and fuse (see 3.3.2.2). Fuse top collarstand interlining to previously fused top collar.						
	(2) Position stay holder to undercollar with side and bottom edges even and stitch along bottom edges to hold stay holder in place.	301	SSv-2	10-14	70-2	70-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. (cont'd)						
	(3) Form a tunnel for the collar stays by stitching the stay holder to undercollar with two rows of stitching 7/16 to 1/2 inch apart according to marks on pattern. The collar points shall be aligned with the center of the two rows of stitching and the stitching shall extend to the raw edges.	301	SSv-2	10-14	70 2	70-2	
	(4) Insert collar stays into the tunnels so tapered ends will abut the top edge stitching when stitched as specified in operation 5.g. Stitch or tack tunnel at bottom of stay holder.	301 or 101	SSv-1 SSv-1	10-14 6-8	70-2 70-2	70-2	
	b. Unslotted collar stay ultrasonically fused construction - (when used).						
	(1) Position fusible interlining to top collar 1/16 inch from top and side edges and fuse (see 3.3.2.2.1). Fuse top collarstand interlining to previously fused top collar.						
	(2) Position collar stays according to marks on pattern so tapered ends will abut the top edge stitching when stitched as specified in operation 5.f.						
	(3) Ultrasonically fuse the collar stays to the undercollar with two rows of fusing points the full length of the stay. Each row shall contain 9 to 11 fusing points per inch and should not cause the collar stay to curl either lengthwise or widthwise. (See 6.8 for recommended ultrasonic welding conditions).						

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					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. (cont'd)						
	c. Slotted collar stay stitched construction- (when used).						
	(1) Position fusible interlining to top collar 1/16 inch from top side edges and fuse (see 3.3.2.2). Fuse top collarstand interlining to previously fused top collar.						
	(2) Position collar stays according to marks on pattern so tapered ends will abut the top edge stitching when stitched as specified in operation 5.g.						
	(3) Stitch collar stays to the undercollar with a 1-1/4 to 1-1/2 inches long tack. The stitching shall straddle the center bar of the stay and not penetrate the plastic material.	101	-	20-24	70-2		
	d. Position top collarstand interlining to top collar fusible as indicated by notches on pattern and with bottom raw edges even, seam along top and sides of stand 1/8 to 3/16 inch from the edge.	301	SSa-1	10-14	70-2	70-2	
	e. With the collar and undercollar face to face, stitch around side and top edge.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	70-2 70-2	70-2 100-2	
	f. Trim, turn, work out edges and points, and press collar points on a collar pointing machine.						
	g. Edge stitch top and sides of collar 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	70-2	70-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make shoulder loops.						
	Finished appearance. The shoulder loops shall finish smooth, flat and uniform in construction and placement. The buttonholes shall be clean cut and correctly positioned. The shoulder loops shall measure finished 2 +0, -1/8 inches wide at the armhole seam tapering to a width of 1-1/2 +0, -1/8 inches at the pointed end.						
	a. Position fusible interlining to top shoulder loop and fuse. Position the shoulder loops face to face with the fused loop on top and stitch the top and under loop around all edges except armhole end.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	70-2 70-2	70-2 100-2	
	b. Turn loop, work out points and edges, and edge stitch along sides and point, 3/16 to 1/4 inch from edge.	301	SSe-2(b)	10-14	70-2	70-2	
	c. Make a horizontal buttonhole in the center of each loop 1/8 inch off-center tolerance, with the inside cut edge of the buttonhole 3/8 to 1/2 inch from point of loop (see 3.7.2).	Buttonhole	Whip or purl types	42-52 per buttonhole including tack	70-2	70-2	
	d. Press shoulder loops with a hot head pressing machine (see operation 22).						
7.	Make pocket flaps.						
	a. Position fusible interlining to top flap and fuse. Position the pocket flaps face to face and seam the flaps along the bottom and sides; turn, work out points and edge stitch 3/16 to 1/4 inch from side and bottom edges.	301 or 401 and 301	SSe-2(a) SSa-2(a) SSe-2(b)	10-14 10-14 10-14	70-2 70-2 70-2	70-2 100-2 70-2	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make pocket flaps. (cont'd)						
	b. Make one vertical buttonhole in the center of each flap (1/8-inch off-center tolerance) with the lower inside cut edge 3/8 to 1/2 inch from bottom edge of flap (see 3.7.2).	Button-hole	Whip or purl types	42-52 per button-hole including tack	70-2	70-2	
	c. Press pocket flaps with a hot head pressing machine (see operation 22).						
8.	Hem breast pocket and make box pleat.						
	a. Fold a vertical box pleat as indicated by marks on pattern to the outside in the center of the pocket and stitch. Pleat shall measure 1-1/4 +1/16 -1/8 inches wide. Press pleat open and flat with a heated pressing iron.	301	OSf-1	10 14	70-2	70 2	
	b. Fold down top edge of pocket as indicated by pattern marks with the raw edge turned under 1/4 inch and stitch 1/16 inch from the folded edge. The hem shall finish 1-1/4 ± 1/8 inches wide.	301 or 401	EFb-1 10-14 EFb-1 10-14		70-2 70-2	70-2 100-2	
	- or -						
	Overedge stitch top raw edge of pocket and turn top of pocket down to form hem. The hem shall finish 1-1/4 ± 1/8 inches wide.	502 or 503	EFd-1 10-14 EFd-1 10-14		100-2 100 2	100-2 100-2	
9.	Make fronts.						
	Superimpose fusible interlining (adhesive side up) to front facing. With front edges even, overedge the fusible interlining to the front of the shirt.	504	SSa-1	14-16	100-2	100-2 or 18 Tex	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make and set pen pocket.						
	a. Fold under top edge of pen pocket as indicated by pattern marks with the raw edge turned under 1/4 inch and stitch 1/16 inch from the folded edge. The hem shall finish 1 ± 1/8 inches wide.	301 or 401	EFb-1 EFb-1	10-14 10-14	70-2 70-2	70-2 100-2	
	- or -						
	Overedge stitch top raw edge of pocket and turn top of pocket down to form hem. The hem shall finish 1-1/4 ± 1/8 inches wide.	502 or 503	EFd-1 EFd-1	10-14 10-14	100-2 100-2	100-2 100-2	
	b. Turn in the side edges of the pen pocket 3/8 inch, and stitch to the left front of the shirt as indicated on the pattern. The finished pen pocket shall be 2-1/4 ± 1/8 inches wide.	301	LSd-1	10-14	70-2	70-2	
	c. Form two compartments with a row of vertical stitching at the center ± 1/8 inch through the shirt front from top to bottom of pocket.	301	SSv-1	10-14	70-2	70-2	
11.	Attach pockets and flaps.						
	Finished appearance. The pocket shall be parallel to the front edge of shirt. The pocket depth at center for all sizes shall finish 5-7/8 ± 3/16 inches, and the finished width shall be 5 ± 3/16 inches wide for all sizes.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Attach pockets and flaps. (cont'd)						
	a. Stitch pockets on fronts, as indicated by marks on patterns, 1/16 to 1/8 inch from edge, continuing stitching 1/4 to 3/8 inch across top of pockets and 1/2 to 5/8 inch diagonally down to the line of 1/16 to 1/8 inch stitching. Do not catch front facing in stitching. The pockets may be precreased.	301 301	LSd-1 and SSv-1	10-14 10-14	100-2 100-2	70 2 70-2	
	b. Press right and left fronts with a hot head pressing machine (see operation 22).						
	c. Position the raw edge of flap on line with marks on patterns and stitch 1/8 to 3/16 inch from raw edge. Turn down flap and raise stitch 3/16 to 1/4 inch from turned edge with the ends of the flap tacked. Pockets and flaps shall be in parallel alignment with each other and front edge of shirt and ends of pocket openings shall be covered by flap. The distance between the top of pocket and the top folded flap edge on the finished shirt shall be 3/4 ± 1/8 inch.	301	LSbl-2	10-14	100-2	70-2	
12.	Attach yoke.						
	Finished appearance. The finished yoke shall be without twists or puckers. There shall be no fullness between the plies of material.						
	a. Position back of shirt between the two plies of yoke and with the back and yoke edges even, stitch all plies together.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 100-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Attach yoke. (cont'd)						
	b. Press back and yoke, with the yoke plies and back in the finished full length position, smooth, flat and even along the stitching with a heated iron at a permanent press (moderate) setting (see operation 22.a.).						
13.	Join shoulder seams.						
	Position fronts between the two plies of yoke and with front and yoke edges even, stitch all plies together. The stitching shall not show on the finished shirt.	301 or 401	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 100-2	
14.	Set shoulder loops.						
	Position shoulder loop so point of loop is 1/4 ± 1/8 inch back of shoulder seam, 3/8 ± 1/8 inch from collar joining finished seam with front edge of loop at armhole opening 1 ± 1/8 inches forward of shoulder seam. Stitch 1/8 to 3/16 inch from raw edge of armhole opening and trim excess even with opening. The excess of the shoulder loop may be trimmed in operation 15.	301 or 401	SSa-1 SSa-1	10-14 10-14	70 2 70-2	70-2 100-2	
15.	Set sleeves.						
	Set sleeve to armhole and join with a safety stitch catching the raw edges of the shoulder loop in the seaming.	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 100 2 (over edge)	100-2 100-2	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Join side and underarm seams.						
	Safety stitch side and sleeve seams in a continuous operation. Armhole seam shall meet at underarm seam, 1/4 inch tolerance.	515 or 516 or 519	SSa-2	10-14	70-2 (chain) and 100-2 (over edge)	100-2 100-2	
17.	Join collar to shirt.						
	Finished appearance. The collar shall be carefully joined to the shirt, with notches matching, without distorting the top or undercollars. Collar ends shall not be uneven in length by more than 1/8 inch. In stitching down top collar, the stitching shall be not more than 1/8 inch on or off the opposite under-collar.						
	a. Fold fronts at notches to form facings. Position the collar assembly between the front facing at the collar notch and stitch from folded edge to within 3/4 ± 1/8 inch of inner edge of facing. Notch collar to neck of shirt.	301	LScf-2(a)	10-14	70-2	70-2	
	b. Turn facings and work out steps. Turn under unsewn portion of top collar and stitch to neck line 1/16 to 1/8 inch from turned edge, catching the ends of the facings in the stitching.	301	LScf-2(b)	10-14	70-2	70-2	
	c. Fuse the folded facing to the shirt body.						
	NOTE: The line of the overedge stitching shall not be visible or cause any impressions on the outside of the shirt after fusing.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Hem shirt bottom.						
	Hem bottom of shirt with the hem extending across fold of fronts. Hem shall finish 1/8 to 5/16 inch wide. The side seams shall be turned in the same direction as sleeve hems.	301	EFb-1	10-14	70-2	70 2	
19.	Hem sleeve.						
	Turn under bottom edge of sleeve, with raw edge turned in, and stitch 1/16 to 1/8 inch from edge. The finished hem shall measure 1 ± 1/8 inch wide. (NOTE: The hemming of sleeve may be performed before or after joining underarm seam.)	301 or 401	EFB-1 EFb-1	10-14 10-14	70-2 70-2	70-2 100-2	
	When sleeve is hemmed prior to joining underarm seams, place a horizontal bartack across seam allowances, at the end of the underarm seam 3/16 to 1/4 inch from bottom of sleeve.	Bartack	--	28 per bartack	70-2	70-2	
20.	Make buttonholes.						
	a. Make six vertical buttonholes in the left front of shirt through the facing as indicated by marks on pattern positioned 3/4 ± 1/16 inch in from front edge (see 3.7.2).	Button-hole	Whip or purl types	42-52 per button-hole including tack	70-2	70-2	
	b. Make one horizontal buttonhole at top on left front of shirt through the facing with front inside cut edge of buttonhole 1/2 ± 1/16 inch from front edge and 3/8 ± 1/16 inch below top of facing (see 3.7.2). The buttonhole shall be parallel to the facing top.	Button-hole	Whip or purl types	42-52 per button-hole including tack	70-2	70-2	

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					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Clean shirts. a. Before folding and packaging, shirts must have all thread ends trimmed and removed, be free of residual fiber particles, lint and dust, or other foreign substances, and have spots and stains removed. b. Remove all shade tickets.						
22.	Pressing. a. As an option to operations 11.b., and 12.b.,a hot head pressing machine capable of pressing the entire shirt after fabrication may be used (see 6.7). b. Press the bottom of shirt, side seams and underarm seams on a hot head pressing machine. NOTE: The use of steam inflated bag type pressing machine is prohibited. The hot head pressing machine cited in operations 6.d., 7.c., and and 11.b., shall be equipped with controls for accurate temperature (350° to 370°F), time (minimum 10 seconds), and pressure (100 psi).						
23.	Sew on buttons. Finished appearance. The buttons shall be aligned and securely sewn in the center of the buttonholes ± 1/16 inch on front of shirt, shoulder loops, and flaps. Collar ends shall be centered equidistant from center of shirt front and shall open 5/8 to 7/8 inch, when collar is buttoned.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Sew on buttons. (cont'd)						
	a. The top button shall be sewn with a back reinforced button stay consisting of basic or alternate material cut as required in operation 1.d. or as per 3.3.1.1. Position the top button with button stay between right front and facing 5/8 ± 1/16 inch from front edge and sew. The stitching shall not penetrate the facing material.	301 or	-	14-16 per button	70-2	70-2	
		101	-	14-16 per button	70-2		
	b. Sew remaining six buttons on right shirt front 3/4 ± 1/16 inch from the front edge through the front facing.	301 or	-	14-16 per button	70-2	70-2	
		101	-	14-16 per button	70-2		
	c. Sew one button on each breast pocket through hem to correspond with the flap buttonhole.	301 or	-	14-16 per button	70-2	70-2	
		101	-	14-16 per button	70-2		
	d. Sew one button on each shoulder seam allowance through yoke to correspond with the buttonhole in the shoulder loop. The shoulder loops shall lay smooth and flat without puckers or twists when they are buttoned.	301 or	-	14-16 per button	70-2	70-2	
		101	-	14-16 per button	70-2		
24.	Touch-up pressing and buttoning shirt.						
	a. When required, touch-up fronts, backs, sleeve hems, and collars, before folding for packaging, with a heated iron at a permanent press (moderate) setting.						
	b. Button shirt fully.						

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3.11 Finished measurements. The finished dimensions for the shirts shall conform to table III.

TABLE III. Finished measurements (in inches)

Shirt size	Neck size 1/	1/2 chest 2/	Back length 3/	Sleeve length 4/	Collar opening 5/
13	13-1/2	18	31	6-1/2	5/8 to 7/8 inch (applicable to all sizes)
13-1/2	14	19	31-1/4	6-1/2	
14	14-1/2	20	31-1/2	6-1/2	
14-1/2	15	21	31-3/4	6-1/2	
15	15-1/2	22	32	6-1/2	
15-1/2	16	23	32-1/4	6-1/2	
16	16-1/2	24	32-1/2	6-1/2	
16-1/2	17	25	32-3/4	6-1/2	
17	17-1/2	26	33	6-1/2	
17-1/2	18	27	33-1/4	6-1/2	
18	18-1/2	28	33-1/2	6-1/2	
Tolerance:					
Plus	1/8	3/8	1/2	1/4	
Minus	1/8	3/8	1/4	3/8	

1/ Neck measurements taken from center of button to front cut-end of buttonhole, with collar in flat position.

2/ Chest measurement taken from folded edge to folded edge at base of armhole with shirt buttoned (see A of figure 1).

3/ Back length measurement taken at center back from collar setting seam to bottom edge of shirt (see B of figure 1).

4/ Sleeve length measurement taken from base of armhole along the inseam to bottom of sleeve (see C of figure 1).

5/ Collar opening measurement shall be taken with the shirt buttoned, placed on a flat surface and without distorting collar, measure across collar from ends of collar at joining seam.

3.12 Workmanship. The finished shirts shall conform to the quality of product established by this specification. As a final step in the contractor's production control plan before formation of a lot, each shirt shall be examined, buttoned, pressed and laid out flat on a table prior to folding for packaging. A shirt containing a selected defect shall not be included in the end item lot.

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Selected defects are those defects listed in MIL-STD-1492, indicated by an asterisk(*). The occurrence of defects shall not exceed the applicable point count value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-1492.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Component testing. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on the components listed in table IV for the characteristics noted. Unless otherwise indicated, testing shall be conducted in accordance with FED-STD-191. All test reports contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size	Sample size (sample units)
Up to 22,000 (inclusive)	8
22,001 and over	13

The basis for lot size and sample units for testing shall be as follows:

Component	Lot size expressed as	Sample unit for testing
Fusible interlining fused to basic cloth	Yards	1 yard full width
Thread	Cones	1 cone

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Fusible interlining fused to basic cloth	Initial adhesion	3.3.2.2.1	5202
	Adhesion after 10 launderings	3.3.2.2.1	5951 1/
Thread	Shrinkage	3.3.4	ASTM D 204 2/
Collar stays	Thickness	3.3.6	5030 3/

1/ Material shall be laundered as specified in 4.5.1.3.1 prior to testing for adhesion.

2/ Shrinkage in dry heat shall not apply.

3/ A certificate of compliance shall be furnished for this characteristic.

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4.4.1.2 Examination of collar stays. Eight collar stays shall be randomly selected from each lot and examined for conformance to configuration and dimensional requirements in 3.3.6. Any nonconforming collar stay found in the sample shall be cause for rejecting the lot represented.

4.4.2 In-process inspection. Visual examination shall be accomplished to ascertain that collar stay tunnels are located and constructed as specified, collar stays are inserted into the collar in the position specified, collar stays are attached and located as specified, and that the fusible material is attached to shirt collar and interlining to shirt fronts, collarstand, and shoulder loops as specified. Evidence of nonconformance shall be cause for rejection of all affected items. Final determination of any suspected nonconformance shall be accomplished at the discretion of the Government by partial or complete disassembly of the end items in question.

4.4.2.1 Adhesion testing of fusibles. Prior to fusing each 1,000 parts, a sample of equivalent area to a part shall be fused for visual and adhesion testing. The fusible shall have a 1 inch starter strip or non-fused area at the top portion of the bond strength samples. This can be accomplished either by placing a strip of paper, folding the fusible onto itself or by placing a thin non-adhesive material between the fusible interlining and outershell before fusing. After visual examination for bubbling, delamination, and bleed through, the sample shall be cut to a 1-inch width and peeled in the warp direction using a spring scale or suitable device capable of measuring the peel strength in ounces. Any sample failing the minimum peel strength requirement specified in 3.3.2.2.3 or any bubbling, delamination or bleed-through shall be cause for rejection of the previous 1,000 fused parts. The rejected lot shall not be used in assembly of the shirts. No fusing shall be performed after the rejection of a 1,000 part lot until the machine has been properly adjusted to produce a passing sample. The rejected lot shall then be replaced with a shade matched lot which has been fused correctly. Rejection of the previously fused parts shall not apply to the first sample tested after machine warm-up. A rejected lot may not be used in the assembly of shirts to be presented to the Government. A rejected lot must be disposed of or set aside with clear markings to avoid accidental use. A rejected lot may be replaced with a lot which has been fused correctly and each part must be properly shade matched to the shirt to which it will be joined. A shaded part defect will be scored on all improperly matched shirts.

4.4.3 End item visual examination. The end items shall be examined for dress shirt (D) visual defects in accordance with the provisions of MIL-STD-1492 except that the defects for knots or slubs (see 6.9) on the outside and for the opening between the collar ends shall be as follows:

Defect	Point value
Knot larger than B, in zone X.	3
Knot larger than C, in zone Y.	1
Slub larger than E in zones X and Y (all areas except collar).	1

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Defect	Point value (cont'd)
Slub larger than C in collar.	3
Collar ends open less than 1/2 inch or more than 1 inch when buttoned.	3

4.4.4 End item dimensional examination. The end items shall be examined for dimensional defects in accordance with the provisions of MIL-STD-1492.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table V. The lot size shall be expressed in units of shirts. The sample unit shall be two shirts. The sample size shall be as specified in 4.4.1.1. One shirt shall be used to evaluate conformance to the appearance requirements and the other shirt shall be used to evaluate for bubbling and delamination of the collar and delamination of the collar stays. The lot shall not be acceptable if one or more sample units fail to meet the requirements for fused collars. Additionally, the lot shall be unacceptable if the lot average (average of all the determinations on the sample units for each characteristic) fails to meet any specified requirements.

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method
Seam appearance rating	3.6.1	4.5.1.2
Fusibles resistance to laundering	3.6.2	4.5.1.1 1/
Ultrasonically fused collar stay resistance to laundering	3.6.3	4.5.1.1 1/

1/ After laundering and conditioning, the fusible parts shall be visually examined for conformance to 3.6.2 and then, when applicable, the collars shall be opened and the stays shall be examined for conformance to 3.7.3.

4.4.6 Packaging examination. The fully packaged end items shall be examined in accordance with the provisions of MIL-STD-1492.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Seam appearance rating (see 6.5) and fusible resistance to laundering test.

4.5.1.1 Laundering procedure.

4.5.1.1.1 Apparatus and material used for laundering. The following apparatus and material is required for laundering:

a. Automatic washing machine (see 6.6).
b. Automatic dryer (see 6.6).
c. AATCC Standard Detergent 124 or equivalent.

4.5.1.1.2 Laundering and conditioning. If the shirts or material selected as test samples weigh less than 4 pounds, ballast to make up a standard 4-pound load shall be added to the test samples. The ballast shall consist of durable press shirts that are approximately the same weight as the test shirts or pieces of durable press fabric that are similar to the fabric of the test samples. The pieces shall be hemmed and measure 3 feet by 3 feet. The front of all shirts in the 4-pound load shall be unbuttoned. The load shall be placed in the washing machine. Add 90 grams of detergent to the washing machine. The washing machine shall be set for 10 minutes on the wash and wear (permanent press) phase, high water level. The washing machine shall be filled with water with a hardness not greater than 50 parts per million (PPM), and the wash water shall be at a temperature of 115° ± 5°F with a cold rinse. Immediately after the washing machine has completed the final spin phase, the entire load shall be removed, untangled, and placed in the dryer. The dryer shall be operated at a temperature of 140° to 160°F until the entire load is dry, continue tumbling 10 minutes with the heat turned off (cool-down phase). Remove the load immediately after the machine stops. The above washing and drying procedure constitutes one (1) complete cycle. Test sample shirts for seam appearance rating shall be subjected to 5 complete cycles. Test sample shirts for bubbling and delamination of collars shall be subjected to 10 complete cycles. Test sample material for adhesion testing shall be subjected to 10 complete cycles. After

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completion of the last cycle, the test sample shall be conditioned for a minimum of 2 hours at 70° ± 10°F at a relative humidity of 50 to 65 percent. (NOTE: During the interruptions between the cycles and during the conditioning and the evaluations, test shirts shall be placed on wooden or plastic hangers and shall be buttoned at the front.)

4.5.1.2 Seam appearance evaluation.

4.5.1.2.1 Apparatus and material used for evaluation. The following apparatus and material is required for evaluation of samples for seam appearance:

a. Overhead lighting procedure apparatus as specified in AATCC Test Method 124-78.

b. AATCC photographic standards for evaluating single needle stitching.

c. Evaluation room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black, or blackout curtains may be hung on each side of the viewing area.

4.5.1.2.2 Evaluation of the appearance of seams. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2, the shirts shall be evaluated by three trained observers. The trained observers, independent of the other observers, shall evaluate each of the shirts as follows: The shirt shall be placed on a clothes hanger with the buttons fastened and the hanger suspended on the viewing board in such a manner that the shirt lies flat on the surface of the board. The photographic standards for single needle stitching shall be placed on the viewing board. The observer shall stand directly in front but 4 feet from the test shirt. The observer shall then compare the test shirt, in each of the areas specified in 3.6.1, to the photographic standards and shall assign a rating value to each of the specified areas. The rating value assigned to an area by the observer shall be the numerical value of the photographic standard that most nearly matches the area. Seams appearing to fall between the numerical values of the photographic standards may be rated 1.5, 2.5, 3.5 or 4.5. Each rating value assigned by each of the observers shall be recorded. The average rating values assigned by each observer to the same area shall be the rating value for that area. If the rating of the same area made by two observers differs by more than 1.0, the rating of all three observers made for that area shall be discarded and the area shall be re-evaluated by 3 different trained observers.

4.5.1.2.3 Bubbling and delamination of collars. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2, the collars shall be examined visually for bubbling and delamination.

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4.5.1.2.4 Adhesion of ultrasonically fused collar stays. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2 and collars examined for bubbling and delamination as specified in 4.5.1.2.3. The collar shall be opened and the stays examined visually for delamination and curling.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each fully buttoned shirt shall be neatly folded to measure approximately 14-1/2 inches by 10-1/2 inches. The folded shirt shall be secured with stainless steel (AISI type 304 or 316) pins, aluminum clips, or plastic fasteners. Each shirt shall be unit packed in a snug-fitting, flat style, clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial preservation. The shirts shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Seventy-two shirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with MIL-STD-1492. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Seventy-two shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety DW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and thirty-six in depth within the shipping container. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with MIL-STD-1492.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with MIL-STD-1492.

5.2.3 Commercial packing. Shirts, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shirts packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The short sleeve shirts are intended for wear by male military personnel of the Department of the Army.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1)
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Seam appearance. In order to avoid puckering and to produce the flattest seams possible, consideration must be given to the following:

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate hole should be no larger than necessary to permit free entry of the needle and to prevent flagging.

c. The finest feed (approximately 18 teeth) possible should be used with a pressure foot to exactly match the shape and size of the feed. The face of the pressure foot (underside next to the feed on the machine) should be highly polished to eliminate roughness and reduce drag. Diamond shaped feed teeth have been used quite successfully.

d. The lightest foot pressure should be used to prevent stretching of the top ply of material.

e. When sewing with stitch type 401 (including the seaming row of the safety stitches), the thread ratio (needle and looper) should be adjusted to produce a stitch with 60 to 70 percent needle thread, and both needle and looper thread tension should be held to the minimum.

f. In the safety stitch machine (515, 516 and 519), the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the end of the fabric.

g. In stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

h. Thread takeup mechanism should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free of burrs, cuts and rough spots.

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6.6 Apparatus. The recommended apparatus for 4.5.1.1 a. and b. and for pressing in operation 22 are:

a. Kenmore Automatic Washing Machine model 600, or similar washing machine. In case of dispute, the Kenmore Model in use at the procurement activity shall be used.

b. Kenmore Automatic Dryer Model 600, or equivalent.

c. "Formatic" hot head pressing machine or equal. "Formatic" manufactured by the American Laundry Machine Co., Cincinnati, OH.

6.7 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table II, providing that the seam and stitch types are as specified and the finished component conforms to the required configurations. When a government finished shaper pattern is furnished, caution shall be taken to assure the component conforms to that pattern.

6.8 Ultrasonic welding conditions. The following ultrasonic welding conditions have been found to provide welds meeting the requirements of this document. These limits are furnished for guidance only. For optimum results with a particular ultrasonic machine used, specific machine settings should be obtained from the manufacturer of the ultrasonic equipment.

	Minimum	Maximum
Pressure, psi	35.0	40.0
Weld time, seconds	0.2	0.3
Down speed, inches/minute	10.0	12.0
Trigger pressure, pounds	8.0	10.0
Hold time, seconds	0.0	0.05

6.9 Defect scale. Sears Fabric Defect Scales are available from:

Fabric Defect Replica Kit
Sears Roebuck and Company
Department 817 (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.10 Subject term (key word) listing.

Clothing, Men's
Garment

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8405-A114)

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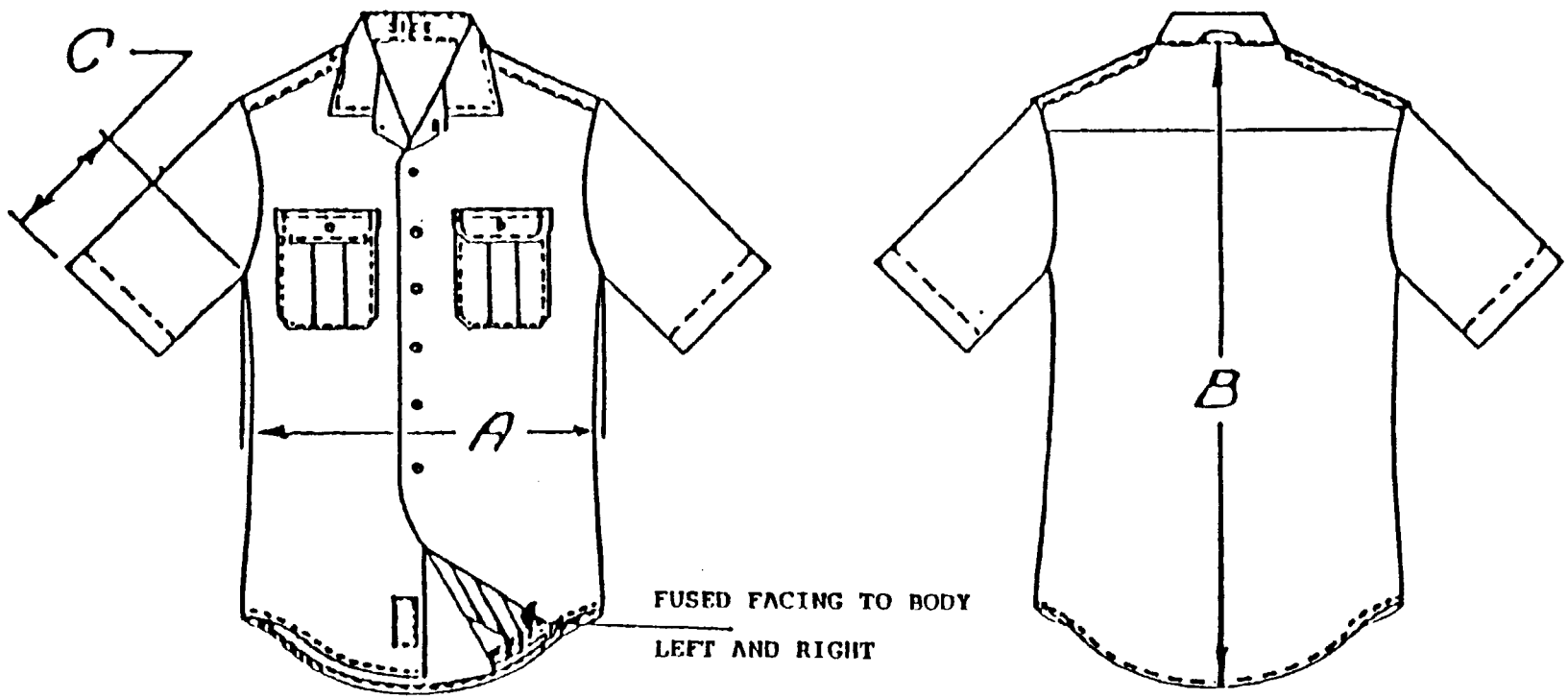


FIG-1 SHIRT, MAN'S, SHORT SLEEVE, ARMY GREEN 415, DURABLE PRESS

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-44041C(GL)	1990 October 9

3. DOCUMENT TITLE
SHIRT, MAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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